

Roll No.

TEE-101

B. TECH. (FIRST SEMESTER)
END SEMESTER EXAMINATION, Jan., 2023
BASIC ELECTRICAL ENGINEERING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define the following terms : (CO1, CO2)

(i) Active and passive elements

(ii) Form factor and peak factor in single phase A. C. circuit

(iii) Thevenin's theorem

(iv) Ideal and practical voltage source

(v) Short circuit and open circuit condition

P. T. O.

- (b) Write down different steps of node voltage analysis. In the network shown in Fig. 1, find the magnitude of each branch current by mesh analysis. (CO1, CO2)

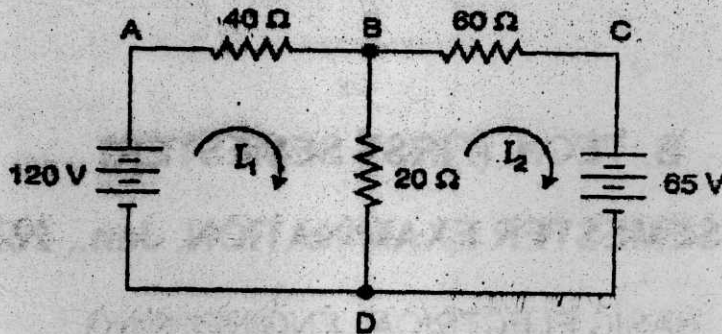


Fig. 1

- (c) State Kirchhoff's voltage law with proper example. For the circuit shown in Fig. 2, find the equivalent resistance across a-b by using star-delta transformation. (CO1, CO2)

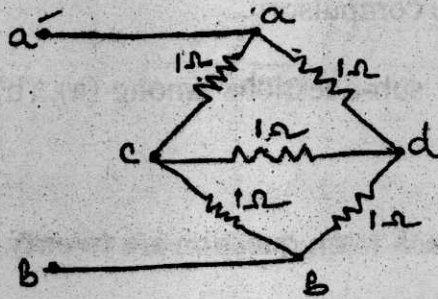


Fig. 2

2. (a) Find the average value, r.m.s. value, form factor, and peak factor of ac voltage whose waveform is shown in Fig. 3. (CO2, CO3)

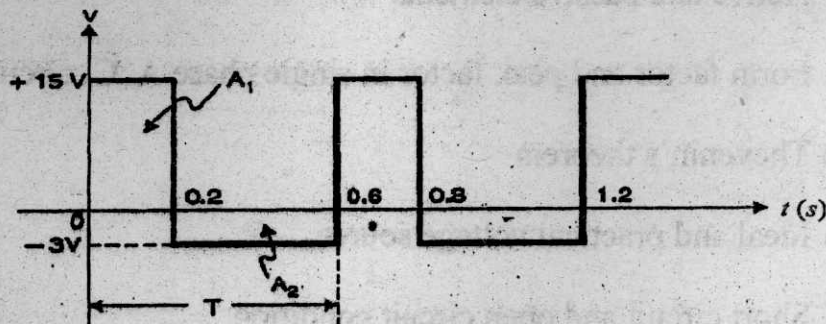


Fig. 3

- (b) Explain the working of single phase purely inductive circuit with proper phasor diagram. Also derive the expression for instantaneous average power. (CO2, CO3)
- (c) Differentiate between electrical wire and cable. Give classification of electrical cable and explain any one. (CO2, CO3)
3. (a) Derive the condition of maximum power transfer in D.C. circuit. Also derive the expression for maximum efficiency in D.C. circuit at maximum power transfer condition. (CO1, CO2, CO4)
- (b) A pure inductance of 318 mH is connected in series with a pure resistance of 75Ω . The circuit is supplied from 50 Hz source and the voltage across 75Ω resistor is found to be 150 V. Calculate the supply voltage and the phase angle. (CO1, CO2, CO4)
- (c) What is the importance of switchgear in electrical system? Classify switchgear on the basis of voltage level. Explain miniature circuit breaker (MCB) in detail. (CO1, CO2, CO4)
4. (a) State Superposition theorem in D. C. circuit. Using superposition theorem, find the current through $1 \text{ k}\Omega$ resistor as shown in Fig. 4. (CO1, CO5, CO6)

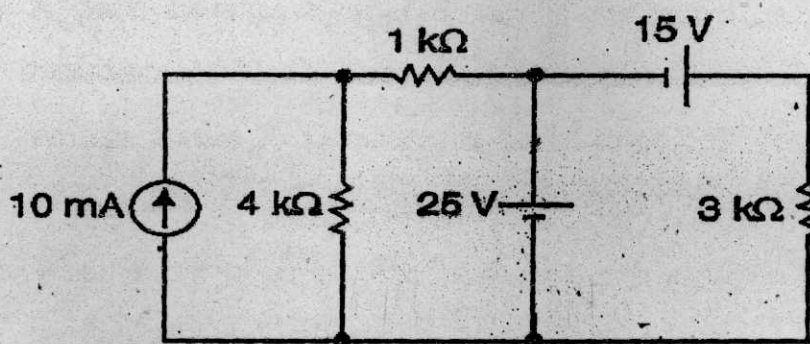


Fig. 4

- (b) What is the need of earthing in electrical system ? How many types of electrical earthing are possible ? (CO1, CO5, CO6)
- (c) An 8-pole wave connected D.C. generator has 1000 armature conductors and flux/pole 0.035 Wb. At what speed it must be driven to generate 500 V. (CO1, CO5, CO6)
5. (a) Define voltage triangle, impedance triangle, and power triangle for single phase series R-L circuit. (CO2, CO6)
- (b) An alternating current i is given by; $i = 141.4 \sin (314 t)$.
Find : (CO2, CO6)
- (i) the maximum value
 - (ii) frequency
 - (iii) time period
 - (iv) the instantaneous value when t is 3 ms.
- (c) Give classification of electrical machines. Also explain the working principle of electrical generator. (CO2, CO6)



Roll No.

TEC-101

**B. TECH. (FIRST SEMESTER)
END SEMESTER EXAMINATION, Jan., 2023**

(All Branches)

BASIC ELECTRONICS ENGINEERING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) If $(73)_x = (54)_y$ and $(54)_x = (40)_y$, find the value of x and y . (CO1)

(b) Solve the following equation with the help of K-map and implement it with NOR gate only : (CO1)

$$F(A, B, C, D) = \sum m(2, 3, 6, 7, 8, 10, 13, 14, 15) + \sum d(0, 1, 5, 12)$$

(c) State and prove De Morgan's theorem. Explain min-terms and max-terms. (CO1)

P. T. O.

2. (a) (i) What is energy band diagram ? Explain classification of materials on the basis of energy band diagram.

(ii) A Si diode has reverse saturation current of $5 \mu\text{A}$ at 300 K . Find forward voltage to achieve a forward current of 1 mA . (CO2)

(b) With the help of a circuit diagram and suitable graph, explain forward bias p - n junction diode. Differentiate between Avalanche breakdown and Zener breakdown. (CO2)

(c) In a germanium sample, an acceptor type impurity is added to the extent of 1 atom per 10^8 Germanium. The density of Ge atoms is $4.41 \times 10^{22}/\text{cm}^3$. Find the electron and hole concentration in the doped semiconductor and calculate the conductivity of germanium as well. Given : (CO2)

$$\mu_n = 3800 \text{ cm}^2/\text{V-sec}$$

$$\mu_p = 1800 \text{ cm}^2/\text{V-sec}$$

$$n_i = 2.5 \times 10^{13}/\text{cm}^3$$

3. (a) Explain the working of half wave rectifier with capacitor filter and draw appropriate circuit diagram as well as input/output waveforms. Also, derive the average and r.m.s. current for half wave rectifier without filter. (CO3)

(b) (i) Explain the working of Zener diode as voltage regulator.

(ii) Define different types of filters. (CO3)

(3)

(c) A full-wave rectifier uses two diodes, the internal resistance of each diode may be assumed constant at $20\ \Omega$. The transformer r.m.s. secondary voltage from centre tap to each of secondary is 50 V and load resistance is $980\ \Omega$. Calculate the following : (CO3)

- (i) I_m, I_{dc}, I_{rms}
- (ii) a.c. power input
- (iii) d.c. power input
- (iv) d.c. output voltage
- (v) efficiency of rectification

4. (a) (i) Derive the relationship between α , β and γ .

(ii) In a transistor if $\alpha = 0.995$ and Emitter current is 10 mA , I_{CBO} is $0.5\ \mu\text{A}$. Then determine I_C, I_B, β and I_{CEO} . (CO4)

(b) Explain the common collector configuration in detail. (CO4)

(c) Draw the common base configuration circuit and explain its current gain, input and output characteristics. (CO4)

5. (a) Derive an expression for output voltage of difference amplifier.

(CO5/CO6)

(b) Derive an expression for output voltage of inverting and non-inverting op-amp. (CO5/CO6)

(c) Derive an expression for output voltage of summing amplifier.

Also design an adder circuit using op-amp. to get $V_o = -(V_1 + 10 V_2 + 100 V_3)$; if $R_F = 100\text{ k}\Omega$. (CO5/CO6)

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TPH-101

B. TECH. (FIRST SEMESTER)
END SEMESTER EXAMINATION, Jan., 2023
ENGINEERING PHYSICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are twenty.

(iv) Each sub-question carries 10 marks.

1. (a) In the wedge-shaped film derive the expression for total path difference. Hence obtain the conditions for maxima and minima. (CO1)
- (b) In N-slit Fraunhofer diffraction using the general expression for the intensity obtain the conditions for principal maxima, minima, and secondary maxima. Hence calculate the intensity of the principal maxima. (CO1)
- (c) A grating having 15000 lines per inch produces spectra of a mercury arc. The green line of the mercury spectrum has a wavelength of 5461 Å. What is the angular separation between the first order and second order green line? (CO1)

P. T. O.

2. (a) What is meant by polarization of light ? Discuss the production of elliptically and circularly polarized light using doubly refracting crystal.
- (b) For the sodium light of wavelength $\lambda = 5890\text{\AA}$, if the velocities of ordinary and extraordinary rays in a calcite crystal are $2.7 \times 10^8 \text{ m/s}$ and $2.95 \times 10^8 \text{ m/s}$ what should be the thickness of the wave plate to produce : (CO2)
- (i) linearly polarized light
- (ii) circularly polarized light ?
- (c) For a two-level system explain the process of stimulated emission. Derive the relation between the Einstein's coefficients. (CO2)
3. (a) Write the postulates for special theory of relativity. Derive the Lorentz transformation equations for space and time coordinates. (CO3)
- (b) Using the Lorentz transformation equations derive the expression for length contraction and time dilation. (CO3)
- (c) Derive the Einstein mass energy relation $E = mc^2$. (CO3)
4. (a) Discuss the various top-down and bottom-up approach in the synthesis of nanoparticles. (CO4, CO5)
- (b) Explain the phenomena of superconductivity. Write the differences between Type I and type II superconductors. Discuss Meissner effect in superconductors. (CO4, CO5)
- (c) Describe the Quantum Well, Wire and Dot nanostructures. Describe in detail with the help of diagram the ball milling method (mechanical method) used in synthesis of nanoparticles. (CO4, CO5)

(3)

5. (a) Derive Schrodinger time dependent and independent wave equation.

(CO6)

- (b) Derive the expression for the eigenvalue of energy of a particle confined in one-dimensional box of infinite height also establish its normalized wave function for the ground level and first excited level. (CO6)

- (c) Find the first three energy levels of an electron moving in one dimension in an infinitely high potential box of width 10^{-8} cm.
(Given mass of electron = 9.11×10^{-28} gm). (CO6)

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TCH-101

B. TECH. (FIRST SEMESTER)
END SEMESTER EXAMINATION, Jan., 2023
ENGINEERING CHEMISTRY

Time : Three Hours

Maximum Marks : 100

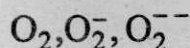
Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Based on molecular orbital diagrams arrange the following molecule/ion in order of their increasing bond length :



Give explanation with proper diagrams. (CO1)

- (b) What is the necessary condition for a molecule to be IR active ? Give the difference between UV-visible and IR spectroscopy. (CO1)

- (c) What do you mean by nanosized materials ? Classify nanomaterials and discuss their applications. (CO1)

P. T. O.

2. (a) Why does hard water consume a lot of soap ? Explain ion exchange method of water softening. Also discuss regeneration of exhausted ion exchanger. (CO2)

(b) Differentiate the following : (CO2)

(i) Ion exchange process and zeolite process

(ii) Cold lime soda process and hot lime soda process

(c) A water sample contains the following impurities per litre :

$\text{Ca}(\text{HCO}_3)_2$ = 16.2 mg, $\text{Mg}(\text{HCO}_3)_2$ = 7.3 mg, CaSO_4 = 6.8 mg,
 MgSO_4 = 12.0 mg, KCl = 200 mg

Calculate temporary and permanent hardness in water. Determine the quantity of lime and soda required in kg for softening 10000 liters of water if the purity of lime and soda are 75% and 90% respectively.

(CO2)

3. (a) What are the applications of conducting polymer with suitable examples. Explain the conductive nature of polymers. (CO3)

(b) Explain the environmental significance of biodegradable polymers. State the preparation and uses of the following types of biodegradable polymers : (CO3)

(i) Biodegradable polymers having ester linkage

(ii) Biodegradable polymers having amide linkage

(c) (i) What are liquid crystal polymers ? Give structure and uses of any one liquid crystal polymer.

(ii) Discuss the following terms by giving suitable examples : (CO3)

(A) Functionality of monomer

(B) Thermoplastics

4. (a) Write characteristics of good fuel and discuss classification of fuels with merits and demerits of each class. (CO4)

(b) Why is net calorific value less than gross calorific value ? Calculate GCV and NCV of a fuel sample from following data : (CO4)

Mass of fuel = 0.95 gm

Water equivalent mass of calorimeter = 650 gm

Amount of water taken in calorimeter = 2.5 kg

Rise in temperature = 1.5°C

Fuse wire correction = 10 cal

Cotton thread correction = 8 cal

Acid correction = 11 cal

% of H = 7%

(c) Differentiate between addition and condensation polymerization with suitable examples. Explain the free radical mechanism of formation of polymers. (CO4)

P. T. O.

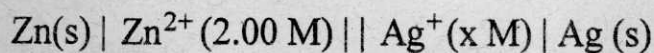
5. (a) What is meant by concentration cell ? Two copper rods are placed in copper sulphate solution of concentration 0.2 M and 0.1 M separately in the form of a cell. Write scheme of the cell and calculate its EMF at 298 K. (CO5)

(b) (i) Differentiate between electrochemical cell and electrolytic cell with examples.

(ii) Derive Nernst's equation and explain the terms involved in it.

(CO5)

(c) A voltaic cell has an E_{cell} value of 1.536 V. What is the concentration of Ag^+ in the cell ? (CO5)



$$\text{Given } E^\circ_{[\text{Zn}^{2+}/\text{Zn}]} = -0.76 \text{ V}; E^\circ_{[\text{Ag}^+/\text{Ag}]} = 0.80 \text{ V}$$

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TCE-101

**B. TECH. (FIRST SEMESTER)
END SEMESTER EXAMINATION, Jan., 2023**

BASIC CIVIL ENGINEERING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are twenty.

(iv) Each sub-question carries 10 marks.

1. (a) Elaborate the contribution of Ancient India in science and technology. (CO1)
(b) Write in detail about any one civil engineering marvel in India. (CO1)
(c) Give the detailed classification of branches of civil engineering. (CO1)
2. (a) Define surveying and discuss the objects of surveying. (CO2)
(b) Distinguish between plane surveying and geodetic surveying. (CO2)
(c) Explain the working of Global Positioning System. (CO2)
3. (a) Draw and label the internal structure of timber. What are the various methods of seasoning of timber? (CO3)

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(2)

- (b) Discuss the various field test of bricks. What are the characteristics of good bricks ? (CO3)
- (c) Discuss the classification of aggregates based on their particle size. What are the various tests mentioned in IS codes for testing of aggregates ? (CO3)
4. (a) Classify the buildings based upon occupancy as per national building code of India. (CO4)
- (b) Mention the factors governing the planning of building. (CO4)
- (c) Enlist the functions of foundation. Discuss the criteria for selection of suitable foundation. (CO4)
5. (a) Enlist some applications of transportation engineering around you and how it is useful in your daily routine. (CO5)
- (b) Write down some water quality parameters along with the standard values assigned to them. If not what will be its impact on human ? (CO5)
- (c) Draw a flow chart to explain how a raw water is treated before it is supplied to the society. (CO5)

Roll No.

TEV-101/22BBGNAET-107/BSCFT-104

**B. TECH./B. B. A./B. SC. (FOOD TECHNOLOGY)
(FIRST SEMESTER)**

END SEMESTER EXAMINATION, Jan., 2023

ENVIRONMENTAL SCIENCE

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are *twenty*.

(iv) Each sub-question carries 10 marks.

1. (a) What are ecological pyramids ? Explain the types of pyramids with examples. (CO1)

(b) What are decomposers ? Discuss the role of decomposers in an ecosystem. (CO1)

(c) Define ecosystem. How energy and nutrient flows in an ecosystem ? (CO1)

2. (a) What is Deforestation ? Discuss various effects of deforestation. (CO4)

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(2)

- (b) Discuss the role of an individual in the conservation of natural resources. (CO14)
- (c) Define Biodiversity. Explain *in-situ* and *ex-situ* conservation of biodiversity. (CO4)
3. (a) Define air pollution. Discuss the sources and effects of air pollution. (CO2, CO5)
- (b) Describe the processes adopted for solid waste management. (CO2, CO5)
- (c) What is water pollution ? What are the effects of water pollution ? (CO2, CO5)
4. (a) Define disaster. Discuss various methods to reduce the impact of flood in detail. (CO2, CO5)
- (b) Write short notes on the following : (CO2, CO5)
- (i) Global Warming
- (ii) Acid rain
- (c) What is sustainable development ? Discuss necessary conditions required to achieve sustainable development. (CO2, CO5)
5. (a) Differentiate between the following : (CO1, CO3)
- (i) Primary and Secondary air pollutants
- (ii) London and Los Angeles smog
- (b) Define food chain and food web. Discuss its significance in an ecosystem. (CO1, CO3)
- (c) Discuss the advantages and disadvantages of dams. (CO1, CO3)

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THU-101

B. TECH. (FIRST SEMESTER)
END SEMESTER EXAMINATION, Jan., 2023
PROFESSIONAL COMMUNICATION

Time : Three Hours

Maximum Marks : 100

- Note :** (i) All questions are compulsory.
- (ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.
- (iii) Total marks in each main question are **twenty**.
- (iv) Each sub-question carries 10 marks.
1. (a) "How we say, what we say." Explain the concept in detail elucidating its components. (CO3)
- (b) Write the components that's are essential while one takes down a message on somebody's behalf. Mention why is it important to put one's signatures towards the end. (CO3)
- (c) Communication is a process that has evolved and gained its current form after a lot of changes done to the first ever way. Elaborate. (CO3)

P. T. O.

2. (a) Write short notes on Constructive and Destructive Feedback: Also justify how negative remarks on something/somebody can be considered as a constructive feedback. (CO4)
- (b) Write short notes on the following : (CO4)
- (i). Why do people complain ?
 - (ii) Complaints as a blessing in disguise.
 - (iii) Write the steps for handling complaints effectively.
- (c) You have reached GD Goenka School, Dehradun for your child's admission in 4th standard. Upon reaching you found that there are 3 more parents waiting in the lounge as the concerned staff has not reached the venue. Pen down a small talk that you initiated with the other 3 parents. (CO4)
3. (a) You are Thomas Edwin, the Manager of a design team in an organization which is into manufacturing dolls. Lately, you observed Emiliano Martínez from Argentina carrying a doll having a baby face of the French player Kylian Mbappe during the Trophy parade. It has been put into your notice that the doll was made in your organization without your knowledge. Write a memorandum, addressing it to Frank Dublin asking the person for showing up with a written explanation about manufacturing such a doll and also about not keeping you informed. (CO5)
- (b) You are Lionel Messi, Captain Soccer Team Argentina. You and your Company have returned to Buenos Aires and for celebrating the victory,

a Trophy Parade is being conducted in the honor of the team. Write a notice to all the players who were a part of FIFA 2022 held at Qatar, to be a part of this parade that starts at 9:30 am on December 20, 2022. Invent any other details if necessary. (CO5)

- (c) You are Lionel Messi and you are chairing a meeting post the parade, on December 21, 2022, wherein the much talked about mocking and trolling of the French player Kylian Mbappe needs to be discussed. Write the agenda of the meeting. Invent necessary details if required.

(CO5)

4. (a) Replace the bold and underlined phrases/words by a suitable word meaning the same.

Post the parade, Messi takes the blame of allowing Emiliano for carrying the doll with Mbappe face on it. Following which he wishes to give up the position of the Captain of the team. Hearing the news all the fans of the hero are in distress and are saying without he as the captain there is little hope from the Team. They are also protesting from getting Messi back and remove the ill-mannered Emiliano from the team instead. They feel that any other captain will not be able to do justice to the sport and all others are young and inexperienced. In spite of all the retaliation, Messi stepped down from the position and Paulo Dybala was crowned as the captain. It was not a team that he could be easily managed by him. He tried to make understand the selectors of his concerns but to no rescue. Finally, he played his first match as the

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captain and made sure to hold the audience spellbound by his performance. After he scored his final goal of the match, he showed intense emotions on the play field, took out Messi's picture from inside his jersey and waved it at his fans. The audience were overwhelmed by amazement and surprise and the faith in the new captain was restored ! (CO2)

- (b) Replace the bold and underlined phrases with a word :

Swami, who considers the happiness of others over his own is the head of the government that is led by one person having all the powers. He owns a place where he keeps all his official documents or records. One day he appointed a man as the in charge of this place. This man had never consumed any intoxicating drink in his life and was a continuous talker during his interview but still could make his place in Swami's team. Though he was a man of such wisdom and presence of mind that could not be overcome by any yet he hated marriages and advised people never to get married as it would ultimately affect their efficiency. One day he met with an accident and upon gaining consciousness he started having loss of memory very frequently. The people who hated him succeeded to injure him critically and left. A man who always wanted to help the strangers in their difficult times, saw him lying on the road and came to his rescue. But, his injuries were deadly, and before dying he uttered his last words in a faint noise which were, "Be nice to people". (CO2)

- (c) (i) His towards his daughter, made the daughter insane with the idea of not being loved. (antipathy, apathy)
- (ii) Mornings in Dehradun witness a lot of (dew/due)
- (iii) His stance made an sense to all, hence he got applauded. (errant, arrant)
- (iv) Keep the pen drive inside the right drawer. (rewrite after correcting the effect of Indianism)
- (v) I greeted him on his second anniversary. (wedding/marriage) (CO2)

5. (a) Choose the correct verb in the following sentences : (CO1)

- (i) This captain, along with all team mates, returns/return to the home land today.
- (ii) The anti-disciplinary committee against Emiliano was/were not convinced to remove him from the team.
- (iii) Either the parents or the baby has/have to get vaccinated.
- (iv) A murder of crows was/were feasting over the dead body.
- (v) The rich flora and fauna of the country need/needs to be preserved.

(b) Fill in the blanks : (CO1)

- (i) He get the property after his father's passing away. (will/shall)
- (ii) You cannot go (unless you have his permission/unless you don't have his permission)

(iii) The baby loves toys. (its/his/her)

(iv) He is than his brother. (richer/more rich)

(v) Choosing Qatar as the organizer of FIFA WC was
choice. (the most ideal/an ideal)

(c) Fill in the correct form of the verb in the following sentences : (CO1)

(i) They playing cricket for 15 years. (be)

(ii) I an inspiring documentary last night. (watch)

(iii) The earth round the sun. (revolve)

(iv) The moon when the climber reached the K2. (shine)

(v) When I left the office this evening, it (already rain)

Roll No.

EEEC-101

B. TECH. (FIRST SEMESTER)

END SEMESTER EXAMINATION, Jan., 2023

BASIC ELECTRICAL AND ELECTRONICS ENGINEERING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are *twenty*.

(iv) Each sub-question carries 10 marks.

1. (a) Define the following terms with suitable example : (CO1, CO2, CO3)

(i) Active and passive elements

(ii) R. M. S. value of an alternating current

(iii) Form factor and peak factor of an ac quantity

(iv) Magneto Motive Force (MMF) in the magnetic circuit

(v) Reluctance in the magnetic circuit

P. T. O.

- (b) State maximum power transfer theorem for a dc circuit. In the circuit shown in Fig. 1, find the current across $R = 20\ \Omega$ by using nodal analysis :
(CO1, CO2, CO3)

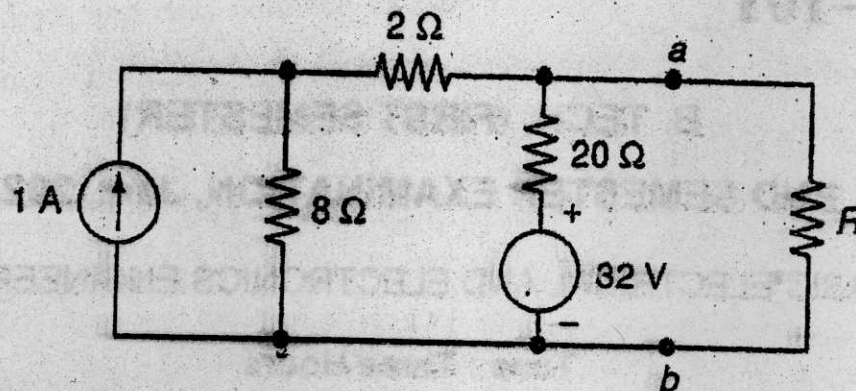


Fig. 1

- (c) State Thevenin's Theorem. Using the superposition theorem, find the current through $1\text{ k}\Omega$ resistor as shown in Fig 2. Assume the sources to be ideal.
(CO1, CO2, CO3)

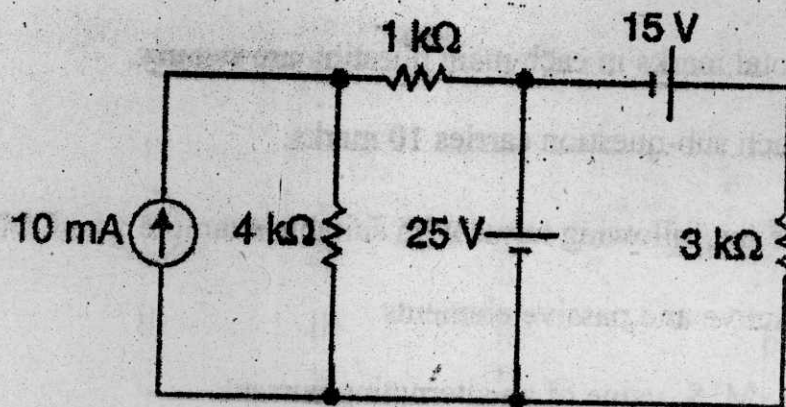


Fig. 2

2. (a) Explain the working of a single phase purely resistive circuit with proper phasor diagram. Also derive the expression for average power.

(CO2)

- (b) Find the average value, r. m. s. value, peak factor, and form factor for half wave rectified alternating current as shown in Fig. 3. (CO2)

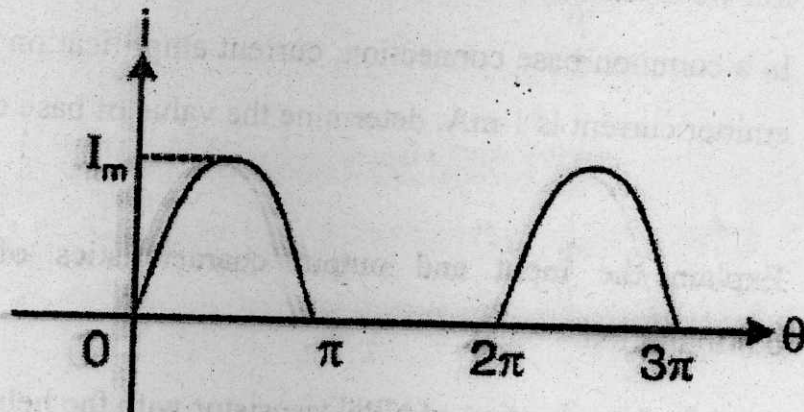


Fig. 3

- (c) Write down the expressions for the current flowing when the voltage, $V = 14.1 \sin(1000 \pi t)$ is applied across : (CO2)
- a 5Ω resistor
 - a 1 mH inductor of negligible resistance and
 - a $150 \mu\text{F}$ capacitor
3. (a) A coil of insulated wire of 500 turns and of resistance 4Ω is closely wound on an iron ring. The ring has a mean diameter of 0.25 m and a uniform cross sectional area of 700 mm^2 . Calculate the total flux in the ring when a d.c. supply at 6 V is applied to the ends of the winding. Assume a relative permeability of 550. (CO3, CO4)
- (b) What are various types of filters used in rectifier circuits ? Draw circuit diagram for each type. (CO3, CO4)
- (c) Explain the working of half wave rectifiers. (CO3, CO4)

4. (a) Find out the average value of load current and DC output voltage for full wave rectifiers. (CO5, CO6)
- (b) In a common base connection, current amplification factor is 0.9. If the emitter current is 1 mA, determine the value of base current. (CO5, CO6)
- (c) Explain the input and output characteristics of common emitter configuration. (CO5, CO6)
5. (a) Explain the operation of NPN transistor with the help of a diagram. (CO6)
- (b) What do you understand by BJT and its types with respect to their doping levels and size ? (CO6)
- (c) Find out the following with respect to NPN transistor : (CO6)
- (i) Current amplification factor
 - (ii) Expression for collector current

Roll No.

TMA-101

B. TECH. (FIRST SEMESTER)
END SEMESTER EXAMINATION, Jan., 2023
ENGINEERING MATHEMATICS—I

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Find the rank of matrix :

(CO1/CO2)

$$A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 4 & 1 & 2 & 1 \\ 3 & -1 & 1 & 2 \\ 1 & 2 & 0 & 1 \end{bmatrix}$$

(b) Find the value of λ for which the system of equations : (CO1/CO2)

$$3x - y + 4z = 3, x + 2y - 3z = -2, 6x + 5y + \lambda z = -3$$

has a trivial solution. Find the solution.

P. T. O.

(c) State's Cayley-Hamilton Theorem and verify it for the matrix :

(CO1/CO2)

$$A = \begin{bmatrix} 5 & -1 & 5 \\ 0 & 2 & 0 \\ -5 & 3 & -15 \end{bmatrix}.$$

2. (a) Find the expansion of $\sin xy$ in powers of $(x-1)$ and $\left(y - \frac{\pi}{2}\right)$ upto quadratic terms. (CO3)

(b) If $y = e^{\tan^{-1} x}$, prove that : (CO3)

$$(1+x^2)y_{n+2} + [2(n+1)x - 1]y_{n+1} + n(n+1)y_n = 0.$$

(c) State's Euler's Theorem and verify it for the function : (CO3)

$$f(x, y) = \sin^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{y}{x}\right).$$

3. (a) If u, v and w are the roots of the equation

$$(\lambda - x)^3 + (\lambda - y)^3 + (\lambda - z)^3 = 0,$$

in λ , then prove that :

(CO3/CO4)

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = -2 \frac{(x-y)(y-z)(z-x)}{(u-v)(v-w)(w-u)}.$$

(b) If $u = xy + yz + zx, v = x^2 + y^2 + z^2, w = x + y + z$, determine whether there is a functional relationship between u, v and w and if so, find it. (CO3/CO4)

(c) Find the extreme values of the function $u(x, y) = x^3 + y^3 - 3axy$.

(CO3/CO4)

(3)

4. (a) Evaluate :

(CO5)

$$\int_0^{\log 2} \int_0^x \int_0^{x+\log y} e^{x+y+z} dz dy dx.$$

(b) Change the order of integration and evaluate :

(CO5)

$$\int_0^\infty \int_0^x x \exp\left(\frac{-x^2}{y}\right) dx dy.$$

(c) Define Beta and Gamma functions and evaluate the integral :

(CO5)

$$\int_0^\infty \frac{x^4(1+x^5)}{(1+x)^{15}} dx.$$

5. (a) Verify Green's Theorem for $\int_C (x^2 - 2xy)dx + (x^2y + 3)dy$ around the boundary C of the region $y^2 = 8x$ and $x = 2$.

(CO6)

(b) Verify Stokes' Theorem for $\vec{F} = x^2\hat{i} - xy\hat{j}$ for the surface integrated around the square in the plane $z = 0$, and bounded by the lines $x = 0$, $y = 0$, $x = a$, $y = a$.

(CO6)

(c) Show that :

(CO6)

$$\text{div}(\text{grad } r^n) = n(n+1)r^{n-2};$$

$$\text{where } r = \sqrt{x^2 + y^2 + z^2}.$$

Roll No.

TCS-101

B. TECH. (FIRST SEMESTER)

END SEMESTER EXAMINATION, Jan., 2023

**FUNDAMENTALS OF COMPUTER AND INTRODUCTION TO
PROGRAMMING**

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question except Question 2.

(iii) Question 2 is compulsory for all.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define Operating System. Explain various functions of operating system. Also describe GUI and CUI. (CO1)

(b) Explain the memory hierarchy. What do you mean by primary and secondary memory ? Differentiate RAM and ROM. (CO1)

P. T. O.

- (c) What do you understand by Computer Network ? Explain different kinds of networks. And also discuss the advantages of Computer Network. (CO1)

2. Find output of following codes : (CO2, CO4, CO5, CO6)

(a) `int main()`

```
{  
    int x=49;  
    for (;x;  
    x--;  
    printf("%d",x);  
    return );  
}
```

(b) `int main( )`

```
{  
    int x;  
    float y;  
    y = x = 7.5;  
    printf("%d %f", x, y);  
    return 0;  
}
```

(3)

TCS-101

(c) int main()

```
{  
    int i = 0, j = 0;  
    while (i < 5, j < 10)  
    {  
        i++;  
        j++;  
    }  
    printf("%d, %d\n", i, j);  
    return 0;  
}
```

(d) int main()

```
{  
    int i = 2;  
    {  
        int i = 4, j = 5;  
        printf("%d %d", i, j);  
    }  
    printf("%d %d", i, j);  
    return 0;  
}
```

P. T. O.

```
(e) int main()  
{  
    int a = 32;  
    do  
    {  
        printf(" %d", a);  
        a++;  
        if(a > 35)  
            break;  
    }while(1);  
}
```

- (f) How many times while loop condition is tested in the following C code snippets, if i is initialized to 0 ?

```
do  
{  
    i++;  
} while (i <= n);
```

```
(g) int main( )  
{  
    int a[4]={1,5};  
    printf("%d",[3]);  
    return 0;  
}
```

(h) void f(int const i)

```
{  
    i=5;  
}
```

int main()

```
{  
    int x = 10;  
    f(x);  
}
```

(i) void solve(int x)

```
{  
    if(x == 0)  
    {  
        printf" %d ", x);  
        return;  
    }  
    printf(" %d ", x);  
    solve(x - 1);  
    printf(" %d ", x);  
}
```

int main()

```
{  
    solve(3);  
    return 0;  
}
```

```
(j) void check( )
{
    int ch = 2;
    switch(ch)
    {
        case 1: printf("1 ");
        case 2: printf("2 ");
        case 3: printf("3 ");
        default: printf("None");
    }
}

int main( )
{
    check( );
    return 0;
}
```

3. (a) Define Following with example :

(CO3, CO4)

- (i) Ladder if-else vs. Switch case
- (ii) Entry controlled loop vs. Exit controlled loop
- (iii) Break vs. Continue
- (iv) Automatic storage class vs. Static storage class

(b) Draw a flowchart to input n positive integers from user. Calculate and print sum of all even numbers and all odd numbers separately.

(CO3, CO4)

- (c) Write a C program to input a positive integer and print it into words.

(CO3, CO4)

Sample Input : 2194

Sample output : Two One Nine Four

4. (a) Define array. Explain various merits and demerits of array. What do you understand by static memory allocation ? (CO3, CO5)
- (b) Draw a flowchart to input n integer elements into an array and replace all elements with their sum of digits. (CO3, CO5)
- (c) Write a C program to input integer elements into a 2D array of size $m \times n$. Find and print minimum element of each row. (CO3, CO5)
5. (a) Explain advantages of functions. What do you understand by function prototype, function calling and function definition with the help of example. (CO6)
- (b) Write a recursive function in C to find sum of following series : (CO6)
 $1 + x^2 + x^4 + x^6 + \dots x^n$ (where n is an even number)
- (c) Write a menu driven program for following calculation and create separate user define function for each case : (CO6)
- (i) Find factorial of an inputted number.
- (ii) Calculate x^y without using `pow()`.

Roll No.

TEV-101/22BBGNAET-107/BSCFT-104

**B. TECH./B. B. A./B. SC. (FOOD TECHNOLOGY)
(FIRST SEMESTER)**

END SEMESTER EXAMINATION, Jan., 2023

ENVIRONMENTAL SCIENCE

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are *twenty*.

(iv) Each sub-question carries 10 marks.

1. (a) What are ecological pyramids ? Explain the types of pyramids with examples. (CO1)
- (b) What are decomposers ? Discuss the role of decomposers in an ecosystem. (CO1)
- (c) Define ecosystem. How energy and nutrient flows in an ecosystem ? (CO1)
2. (a) What is Deforestation ? Discuss various effects of deforestation. (CO4)

P. T. O.

(2)

- (b) Discuss the role of an individual in the conservation of natural resources. (CO14)
- (c) Define Biodiversity. Explain *in-situ* and *ex-situ* conservation of biodiversity. (CO4)
3. (a) Define air pollution. Discuss the sources and effects of air pollution. (CO2, CO5)
- (b) Describe the processes adopted for solid waste management. (CO2, CO5)
- (c) What is water pollution ? What are the effects of water pollution ? (CO2, CO5)
4. (a) Define disaster. Discuss various methods to reduce the impact of flood in detail. (CO2, CO5)
- (b) Write short notes on the following : (CO2, CO5)
- (i) Global Warming
- (ii) Acid rain
- (c) What is sustainable development ? Discuss necessary conditions required to achieve sustainable development. (CO2, CO5)
5. (a) Differentiate between the following : (CO1, CO3)
- (i) Primary and Secondary air pollutants
- (ii) London and Los Angeles smog
- (b) Define food chain and food web. Discuss its significance in an ecosystem. (CO1, CO3)
- (c) Discuss the advantages and disadvantages of dams. (CO1, CO3)